

Question ID: bc7b1a04

Question

| Three Studies' Estimated Average Velocity of LMC |            |                            |
|--------------------------------------------------|------------|----------------------------|
| Researchers                                      | Study year | Estimated average velocity |
| Murai and Fujimoto                               | 1980       | 344 km/s                   |
| Kallivayalil and colleagues                      | 2006       | 378 km/s                   |
| Gardiner and colleagues                          | 1994       | 297 km/s                   |

In 2006, Nitya Kallivayalil and colleagues calculated the most accurate estimate yet of the average velocity (in kilometers per second) of the Large Magellanic Cloud (LMC) galaxy. Before the 2006 study, estimates of the average velocity were low enough for the LMC to maintain an orbit around the Milky Way galaxy, but according to an analysis by Gurtina Besla and colleagues, the estimated velocity from the 2006 study is too high for the LMC to maintain such an orbit. Therefore, if Besla and colleagues are correct, the maximum average velocity for the LMC that would allow it to maintain orbit around the Milky Way is likely \_\_\_\_\_

Which choice most effectively uses data from the table to complete the statement?

Answer

- A. above 344 km/s but below 378 km/s.
- B. above 297 km/s but below 344 km/s.
- C. above 378 km/s.
- D. below 297 km/s.